

Threshold Tomography Reconstruction of a Latent 5D Curve

1 Problem Setup

Given binary observations

$$\chi(s) \in \{0, 1\},$$

a scan trajectory $S(s)$, and an unknown latent curve

$$\gamma(t) \subset \mathbb{R}^5,$$

recover $\gamma(t)$ from thresholded regime-switch measurements.

2 Forward Model

Define the distance from a scan point to the curve:

$$h(s, \gamma) = \min_t \|S(s) - \gamma(t)\|.$$

Using a soft threshold approximation,

$$\hat{\chi}(s; \gamma) = \sigma(k(\epsilon - h(s, \gamma))),$$

where σ is the logistic sigmoid.

3 Curve Representation

Represent the curve by control points

$$\gamma \approx \{g_1, \dots, g_N\}, \quad g_i \in \mathbb{R}^5.$$

Use a smoothness penalty

$$R(\gamma) = \sum_{i=2}^{N-1} \|g_{i+1} - 2g_i + g_{i-1}\|^2.$$

4 Inverse Problem

Minimize

$$J(\gamma) = \sum_i (\chi_i - \hat{\chi}_i)^2 + \lambda R(\gamma).$$

5 Reconstruction Algorithm

1. Initialize γ from the positive detections ($\chi = 1$).
2. Compute distances $h(s_i, \gamma)$.
3. Predict $\hat{\chi}_i$.
4. Compute gradients of J with respect to the control points.
5. Update control points via gradient descent.
6. Refit a principal curve through the positive-hit set.
7. Repeat until convergence.

6 Interpretation

This is a manifold reconstruction problem rather than classical density tomography. The reconstruction estimates a low-dimensional object from thresholded observations of its tubular neighborhood.